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Image Enhancement in Spatial Domain

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Image enhancement in the spatial domain improves the visual quality of an image by directly modifying the intensity (pixel) values.

① gray-level transformation

Gray level transformation changes the intensity values of pixels.

Types

=> Image Negative

=> Low Transformation

=> Power-law

② Histogram processing.

A histogram shows the distribution of pixel intensities.

Techniques :-

=> histogram Equalization

=> histogram specification

=> histogram stretching

③ Spatial filtering :-

Spatial filtering uses a small mask (kernel) that move across the image.

Types

→ Smoothing filters

→ Sharpening filters

Difference Between Smoothing and Sharpening

Image smoothing

Removes noise

Blurs image slightly

uses - low-pass filter

Reduces intensity variation

Image sharp

Enhances edge

makes image

uses high-pass

increase intens

image smoothing

= A Smoothing filter replaces each pixel with the average or median value of neighboring pixels

Types :-

mean filter

Gaussian filter

Median filter

Advantages :-

* Remove salt and pepper noise

* produces smooth images

Image Sharpening:-

=> Sharpening emphasize rapid intensity changes by highlighting edges.

Types:-

- * Laplacian filter
- * Sobel filter
- * Prewitt filter

Advantages:-

- * highlights object boundaries
- * improves features visibility

Frequency domain enhancement

Instead of modifying pixels directly the image is transformed into the frequency domain using Fourier transform.

Steps:-

- => convert image to frequency domain
- => Apply filter
- => perform inverse Fourier transform
- => obtain enhanced image.

Low-Pass filter (LPF)

Allows low frequency to pass and block high frequencies.

effects

=> Remove noise => produces smooth image

=> Reduces sharp edges.

Applications :-

=> Medical imaging => Noise removal

=> satellite images.

high-pass filter (HPF)

Allows high frequencies and blocks low frequencies.

effects :-

-> enhance edge => Highlights fine details

=> increase sharpness.

Applications :-

=> Fingerprint recognition

=> OCR

=> industrial inspection

Comparison

Technique	Working Principle	Advantages	Limitation	Application
Point Detection	Points isolated purely	Simple	Detects only points	Defect detection
Line Detection	Detects horizontal or vertical	Detects structures	Sensitive to noise	Read detection
Edge Detection	Detects sudden intensity	Accurate boundaries	Noise sensitive	False detection.
Thresholding	Separates foreground	Fast	Poor under uneven	Document Scanning
Region-Based Segmentation	Group similar neighbouring	Accurate Segmentation	Computationally expensive	Medical imaging

relation sudden changes in pixel intensity

- * methods: Sobel, Prewitt, Canny

Edge linking :-

- * connects broken edges to form continuous edges.
- * uses edges continuity and connectivity

Boundary Detection :-

- * Boundary detection identifies the complete outer contour of an object
- * uses linked edges to create closed boundaries